CCNA / CCNP QUICK REFERENCE

Cisco IOS Commands Cheat Sheet

Comprehensive CLI Guide with Domain Name, SSH v2, User Auth, Auto-MDIX, Interface Diagnostics, VLAN Deletion / Factory Reset, EtherChannel, Static & Floating Routes, Switch Port Security, DHCPv4 & DHCPv6

EXEC MODES & NAVIGATION

User EXEC Mode	`Router>`
Privileged EXEC Mode	`enable`
Global Configuration	`configure terminal`
Exit Mode / End Config	`exit / end`
Context Help	`? or command ?`

DOMAIN, SSH V2 & USER AUTHENTICATION

1. Set Hostname	`hostname <name>`
2. Set IP Domain Name	`ip domain-name <domain.com>`
3. Generate RSA Keys	`crypto key generate rsa general-keys modulus 2048`
4. Enforce SSH Version 2	`ip ssh version 2`
5. Create Local User	`username <user> secret <pass>`
6. Enable Local Auth on VTY line	`vty 0 4` `login local`
7. Restrict VTY to SSH transport	`transport input ssh`
Verify SSH Status	`show ip ssh | show ssh`

TIP — Security Best Practice

Always enforce SSH Version 2 (`ip ssh version 2`) as SSHv1 contains known security vulnerabilities. Generating an RSA key modulus of at least 2048 bits is required for SSHv2.

GLOBAL CONFIG & SECURITY

Encrypted Enable Secret	`enable secret <pass>`
Console Port Password	`line con 0` `password <pass> | login`
Encrypt Passwords	`service password-encryption`
Disable IP Domain Lookup	`no ip domain-lookup`
Save Configuration	`copy run start`

INTERFACE SPEED, DUPLEX & AUTO-MDIX

Select Interface / Range	`interface fa0/1` `interface range fa0/1 - 12`
Configure Port Speed	`speed auto | 10 | 100 | 1000`
Configure Duplex Mode	`duplex auto | full | half`
Enable Auto-MDIX	`mdix auto`
Enable / Shutdown Port	`no shutdown | shutdown`

NOTE — Auto-MDIX Rule

Automatic MDIX cable detection requires both `speed auto` and `duplex auto` to be active on the interface.

VLANS & TRUNKING

Create / Name VLAN	`vlan <id>` `name <vlan_name>`
Assign Access VLAN	`switchport mode access` `switchport access vlan <id>`
Assign Voice VLAN	`switchport voice vlan <id>`
Trust QoS CoS Markings (Voice)	`mls qos trust cos`
Remove VLAN Assignment from Port	`no switchport access vlan`
Configure Trunk Port	`switchport mode trunk`
Allowed Trunk VLANs	`switchport trunk allowed vlan <list>`

TIP — 'do' Command

Use the do prefix to run privileged EXEC commands (e.g. do `show vlan brief`) without leaving configuration mode.

ETHERCHANNEL (LACP / PAGP)

Select Member Interfaces	`interface range fa0/1 - 2`
Bundle into Port-Channel (LACP)	`channel-group <id> mode active | passive`
Bundle into Port-Channel (PAgP)	`channel-group <id> mode desirable | auto`
Bundle into Port-Channel (Static)	`channel-group <id> mode on`
Select Port-Channel Interface	`interface port-channel <id>`
Set Port-Channel as Trunk	`switchport mode trunk`
Allowed VLANs on Port-Channel	`switchport trunk allowed vlan <list>`
Remove a Member from the Channel	`no channel-group`

View Channel Summary	`show etherchannel summary`
View Port-Channel Details	`show etherchannel <id> port-channel`
View Port Status in Channel	`show etherchannel port`
View LACP Neighbor Info	`show lacp neighbor`
View PAgP Neighbor Info	`show pagp neighbor`

NOTE — Negotiation Protocols

LACP (IEEE 802.3ad, open standard) uses active / passive — at least one side must be active. PAgP (Cisco proprietary) uses desirable / auto — at least one side must be desirable. `mode on` forces a static EtherChannel with no negotiation on either side. Mixing incompatible modes on the two ends (e.g. LACP active against PAgP desirable) prevents the channel from forming.

Rule: All member ports must match before bundling — same speed, duplex, trunk/access mode, allowed VLANs, and native VLAN. Mismatched settings keep ports out of the bundle or put the channel in an error/suspended state.

Example — LACP EtherChannel Trunk:

```
S1(config)# interface range fa0/1 - 2
S1(config-if-range)# channel-group 1 mode active
S1(config-if-range)# exit
S1(config)# interface port-channel 1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk allowed vlan 10,20,99
```

ROUTER-ON-A-STICK (TRUNK SUBINTERFACES)

Select Subinterface	`interface <phys_intf>.<subintf_id>` `e.g. interface G0/0/1.10`
Add Description	`description <text>`
Tag VLAN (802.1Q Encapsulation)	`encapsulation dot1Q <vlan_id>`
Assign Gateway IP Address	`ip address <ip> <subnet_mask>`
Return to Global Config	`exit`

Enable Physical Trunk Interface	`interface <phys_intf>`
	`description Trunk link to <switch>`
	`no shutdown`

NOTE — Router-on-a-Stick

Create one subinterface per VLAN on the router's physical interface connected to the switch trunk port. Each subinterface is tagged with `encapsulation dot1Q <vlan_id>` and given an IP address that serves as the default gateway for that VLAN. The parent physical interface carries no IP address of its own and must be enabled with `no shutdown`.

Example — 3 VLANs on G0/0/1:

```
interface G0/0/1.10
description Default Gateway for VLAN 10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit
interface G0/0/1.20
description Default Gateway for VLAN 20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
interface G0/0/1.99
description Default Gateway for VLAN 99 (Native/Mgmt)
encapsulation dot1Q 99
ip address 192.168.99.1 255.255.255.0
exit
interface G0/0/1
description Trunk link to S1
no shutdown
```

MAC ADDRESS TABLE MANAGEMENT

Add Static MAC Address	`mac address-table static <mac_addr> vlan <id> interface <intf>`
Remove Static MAC Address	`no mac address-table static <mac_addr> vlan <id>`
Set Dynamic Aging Time	`mac address-table aging-time <seconds>`
Disable MAC Aging (per VLAN)	`mac address-table aging-time 0 vlan <id>`
Clear Dynamic Entries (All)	`clear mac address-table dynamic`
Clear Dynamic Entries (Interface)	`clear mac address-table dynamic interface <intf>`
Clear Dynamic Entries (VLAN)	`clear mac address-table dynamic vlan <id>`
Show Full MAC Address Table	`show mac address-table`
Show Static Entries Only	`show mac address-table static`
Show Dynamic Entries Only	`show mac address-table dynamic`
Show Entries for a VLAN	`show mac address-table vlan <id>`
Show Entries for an Interface	`show mac address-table interface <intf>`
Show Current Aging Time	`show mac address-table aging-time`

NOTE — Static MAC Addresses

A static entry permanently binds a specific host's MAC address to a switch port and VLAN, so that host is only ever learned on that port — useful for servers, printers, or security-sensitive devices. Static entries do not age out and survive normal MAC table refreshes, but are lost on reload unless the configuration is saved with `copy run start`.

Example:

```
S1(config)# mac address-table static 000A.000B.000C vlan 10 interface fa0/6
S1(config)# do show mac address-table static
```

SWITCH PORT SECURITY & HARDENING

NEW

Enable Port Security	`switchport port-security`
Set Maximum MAC Addresses	`switchport port-security maximum <1-N>`
Learn MACs Dynamically (Sticky)	`switchport port-security mac-address sticky`
Manually Bind a MAC Address	`switchport port-security mac-address <mac_addr>`
Set Violation Action	`switchport port-security violation protect | restrict | shutdown`
Set Aging Time / Type	`switchport port-security aging time <minutes>` `switchport port-security aging type absolute | inactivity`
Verify Port Security Summary	`show port-security`
Verify Per-Interface Detail	`show port-security interface <intf>`
Verify Learned/Sticky Addresses	`show port-security address`
Re-enable an Err-Disabled Port	`shutdown` `no shutdown`
Auto-Recover Err-Disabled Ports	`errdisable recovery cause psecure-violation` `errdisable recovery interval <seconds>`

Disable Unused / Unassigned Ports	`interface range fa0/10 - 24`
	`shutdown`
	`switchport mode access`
	`switchport access vlan <unused_id>`

Enable BPDU Guard (Access Ports)	`spanning-tree portfast` `spanning-tree bpduguard enable`
Enable Root Guard (Uplink Ports)	`spanning-tree guard root`
Enable DHCP Snooping (Global)	`ip dhcp snooping` `ip dhcp snooping vlan <list>`
Trust Uplink for DHCP Snooping	`ip dhcp snooping trust`
Rate-Limit DHCP on Access Port	`ip dhcp snooping limit rate <pps>`
Enable Dynamic ARP Inspection	`ip arp inspection vlan <list>`
Trust Uplink for ARP Inspection	`ip arp inspection trust`
Set Login Attempt Rate Limiting	`login block-for <seconds> attempts <n> within <seconds>`
Set an Unauthorized-Access Banner	`banner motd # <message> #`

NOTE — Violation Modes

`protect` — drops offending traffic silently, no log/counter increment. `restrict` — drops offending traffic, increments the violation counter, and sends an SNMP/syslog notification. `shutdown` (default) — puts the port into **err-disabled** state, taking it completely down until manually or automatically recovered.

CAUTION — Manually Recovering an Err-Disabled Port

A port placed in **err-disabled** state by a security or violation event stops forwarding all traffic. Unless automatic recovery is configured with `errdisable recovery cause psecure-violation`, the port must be manually cycled with `shutdown` followed by `no shutdown` to restore connectivity.

TIP — Layered Access-Edge Hardening

Port security, DHCP snooping, Dynamic ARP Inspection (DAI), and BPDU/Root Guard are complementary, not redundant. DHCP snooping builds the trusted binding table that DAI relies on to validate ARP replies, while BPDU Guard and Root Guard protect the spanning-tree topology on access and uplink ports respectively. Apply all of them together at the access layer for defense in depth.

Example — Hardened Access Port with Sticky Port Security:

```
S1(config)# interface fa0/6
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
S1(config-if)# switchport port-security
S1(config-if)# switchport port-security maximum 2
S1(config-if)# switchport port-security mac-address sticky
S1(config-if)# switchport port-security violation restrict
S1(config-if)# spanning-tree portfast
S1(config-if)# spanning-tree bpduguard enable
S1(config)# errdisable recovery cause psecure-violation
S1(config)# errdisable recovery interval 300
```

VLAN DELETION & FACTORY RESET

Remove a Single VLAN	`no vlan <vlan-id>`
Delete Entire vlan.dat File	`delete flash:vlan.dat`
Delete vlan.dat (abbreviated, default location only)	`delete vlan.dat`
Erase Startup Config	`erase startup-config`
Reload Switch	`reload`

CAUTION — Reassign Ports Before Deleting a VLAN

The `no vlan <vlan-id>` global configuration mode command removes a VLAN from the switch's vlan.dat file. Before deleting a VLAN, reassign all member ports to a different VLAN first. Any ports that are not moved to an active VLAN are unable to communicate with other hosts after the VLAN is deleted, and will remain unable to do so until they are assigned to an active VLAN.

NOTE — VLAN.DAT Deletion — Factory Default VLAN Config

The entire vlan.dat file can be deleted using the `delete flash:vlan.dat` privileged EXEC mode command. The abbreviated command version (`delete vlan.dat`) can be used if the vlan.dat file has not been moved from its default location. After issuing this command and reloading the switch, any previously configured VLANs are no longer present — this effectively places the switch into its factory default condition with regard to VLAN configurations.

NOTE — Full Factory Reset Procedure

To restore a Catalyst switch to its factory default condition: unplug all cables except the console and power cable from the switch, then enter the `erase startup-config` privileged EXEC mode command, followed by the `delete vlan.dat` command. Reload the switch to complete the reset.

VERIFICATION & DIAGNOSTICS (SHOW)

show ip interface brief	IP & status summary for all interfaces.
show interfaces fa0/1	Link status, speed, duplex, & error counters.
show interfaces status	Switchport mode, speed, duplex, & VLAN.
show controllers ... phy | inc MDIX	Verifies if Auto-MDIX is On or Off.
show running-config	Displays active running configuration in RAM.
show vlan brief	Lists all active VLANs and their assigned ports.
show vlan name <vlan_name>	Shows VLAN details filtered by VLAN name.
show vlan id <vlan_id>	Shows VLAN details filtered by VLAN ID number.
show vlan summary	Displays a count of existing, VTP, & extended VLANs.
show interface <intf> switchport	Detailed switchport mode, access VLAN, & voice VLAN info.
show mac address-table	MAC-to-port binding table.
clear counters fa0/1	Resets error counters on port Fa0/1.

MEDIA ERROR TROUBLESHOOTING

Late Collisions (> 0)	Duplex mismatch (one end Full, one end Half) or Ethernet cable exceeding 100m.
CRC Errors (> 0)	Physical interference, faulty copper cabling, bad patch panel, or loose RJ-45 pins.
Runts (> 0)	Collisions on half-duplex links or a malfunctioning host NIC sending undersized frames (< 64 bytes).
Interface UP / Protocol DOWN	Layer 2 issue – speed/duplex mismatch, encapsulation, or VLAN assignment error.

IP ROUTING (STATIC & OSPF)

Static Route	`ip route <net> <mask> <next_hop>`
Default Route	`ip route 0.0.0.0 0.0.0.0 <next_hop>`
Enable OSPF	`router ospf <process_id>`
OSPF Network Statement	`network <ip> <wildcard> area <id>`

STATIC & FLOATING STATIC ROUTES

NEW

Standard Static Route (AD 1)	`ip route <net> <mask> <next_hop>`
Static Route via Exit Interface	`ip route <net> <mask> <exit_intf>`
Static Route with Explicit AD	`ip route <net> <mask> <next_hop> <AD>`
Floating Static Route (Backup Path)	`ip route <net> <mask> <backup_next_hop> <AD>`
Floating Default Route	`ip route 0.0.0.0 0.0.0.0 <backup_next_hop> <AD>`

Track Route Reachability (SLA)	`ip sla <id>`
	`icmp-echo <target_ip> source-interface <intf>`
	`frequency <seconds>`

Schedule the SLA Operation	`ip sla schedule <id> life forever start-time now`
Create a Track Object from SLA	`track <id> ip sla <sla_id> reachability`
Tie a Static Route to a Track Object	`ip route <net> <mask> <next_hop> track <track_id>`
View Routing Table	`show ip route`
View Static Routes Only	`show ip route static`
View a Specific Route/Network	`show ip route <net>`
View Track Object Status	`show track <id>`
View IP SLA Statistics	`show ip sla statistics`

NOTE — What Makes a Route 'Floating'

A floating static route is simply a static route configured with an **Administrative Distance (AD)** higher than the primary route it is meant to back up — for example higher than a standard static route (AD 1), OSPF (AD 110), or EIGRP (AD 90). Because IOS always prefers the route with the **lowest** AD for a given destination, the floating route stays out of the routing table entirely while the primary route is up, and only 'floats' into the table automatically when the primary route disappears.

TIP — Object Tracking for Faster Failover

Tying a floating static route to a tracked object (via IP SLA + `track`) lets IOS pull the primary route the instant the tracked destination becomes unreachable — rather than waiting on a routing protocol's own convergence timers. This is the standard method for fast WAN/ISP failover on a router with dual uplinks.

Example — Floating Static Route as ISP Backup:

```
! Primary route learned via OSPF (AD 110) — no static route configuration needed here.
! Floating static default route as backup, only used if the OSPF route is lost:
R1(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.2 200
! Optional — verify only the primary route is active while it is up:
R1# show ip route static
```

Example — Floating Static Route with Object Tracking:

```
R1(config)# ip sla 1
R1(config-ip-sla)# icmp-echo 198.51.100.1 source-interface g0/0
R1(config-ip-sla-echo)# frequency 5
R1(config-ip-sla-echo)# exit
R1(config)# ip sla schedule 1 life forever start-time now
R1(config)# track 1 ip sla 1 reachability
R1(config)# ip route 0.0.0.0 0.0.0.0 192.168.1.1 track 1
R1(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.2 200
```

FIRST HOP REDUNDANCY PROTOCOLS (HSRP / VRRP / GLBP) NEW

Assign HSRP Virtual IP	`interface <intf>` `standby <group> ip <virtual_ip>`
Use HSRP Version 2	`standby version 2`
Set HSRP Priority (Higher Wins)	`standby <group> priority <0-255>`
Enable HSRP Preemption	`standby <group> preempt`
Tune HSRP Hello/Hold Timers	`standby <group> timers <hello_sec> <hold_sec>`
Set HSRP MD5 Authentication	`standby <group> authentication md5 key-string <key>`
Track an Interface (HSRP)	`standby <group> track <intf> decrement <value>`
Assign VRRP Virtual IP	`interface <intf>` `vrrp <group> ip <virtual_ip>`
Set VRRP Priority (Higher Wins)	`vrrp <group> priority <1-254>`
Enable VRRP Preemption (Default On)	`vrrp <group> preempt`
Tune VRRP Advertisement Timer	`vrrp <group> timers advertise <seconds>`
Track an Interface (VRRP)	`vrrp <group> track <intf> decrement <value>`
Assign GLBP Virtual IP	`interface <intf>` `glbp <group> ip <virtual_ip>`
Set GLBP Priority (Higher Wins AVG)	`glbp <group> priority <0-255>`
Enable GLBP Preemption	`glbp <group> preempt`
Set GLBP Load-Balancing Method	`glbp <group> load-balancing round-robin | weighted | host-dependent`

Track an Interface (GLBP)	`glbp <group> weighting track <object_id> decrement <value>`
View HSRP Status	`show standby brief` `show standby`
View VRRP Status	`show vrrp brief` `show vrrp`
View GLBP Status	`show glbp brief` `show glbp`

NOTE — How FHRP Election Works

All three First Hop Redundancy Protocols let two or more routers share a single virtual IP (and virtual MAC) as the LAN's default gateway, so a router failure doesn't take down host connectivity. **HSRP** (Cisco proprietary) elects one *Active* and one *Standby* router per group — the highest priority wins (default 100), ties broken by highest real IP address. **VRRP** (open standard, RFC 5798) works the same way but calls the winner the *Master* and the loser(s) *Backup*. **GLBP** (Cisco proprietary) elects an *Active Virtual Gateway* (AVG) that in turn assigns multiple *Active Virtual Forwarders* (AVFs), allowing several routers to forward traffic simultaneously for true load balancing rather than a single active path.

TIP — Preempt & Priority

A higher-priority router does **not** automatically reclaim the Active/Master role after recovering from a failure unless `preempt` is explicitly enabled — VRRP enables preemption by default, while HSRP and GLBP do not. Combine preemption with interface or object tracking (`standby track`, `vrrp track`, `glbp weighting track`) so priority automatically drops — and the standby router takes over — if a router's uplink goes down, not just if the router itself fails.

NOTE — Version & Group Ranges

HSRPv1 (default) supports groups 0–255 and uses multicast 224.0.0.2; HSRPv2 (`standby version 2`) supports groups 0–4095 and uses multicast 224.0.0.102 — both routers on a link must run the same version. VRRP and GLBP each support groups 0–255.

Example — HSRP Active/Standby Pair with Tracking:

```
! R1 – intended Active router
R1(config)# interface g0/0
R1(config-if)# standby version 2
R1(config-if)# standby 1 ip 192.168.1.1
R1(config-if)# standby 1 priority 150
R1(config-if)# standby 1 preempt
R1(config-if)# standby 1 track g0/1 decrement 60
! R2 – Standby router
R2(config)# interface g0/0
R2(config-if)# standby version 2
R2(config-if)# standby 1 ip 192.168.1.1
R2(config-if)# standby 1 priority 100
```

IPV4 DHCP (SERVER, POOL & RELAY)

Exclude Address(es) from Pool	`ip dhcp excluded-address <low> [high]`
Create DHCP Pool	`ip dhcp pool <pool-name>`
Define Network / Subnet	`network <net-ip> <subnet-mask | /prefix>`

Set Default Gateway(s)	`default-router <ip-1> [ip-2 ... ip-8]`
Set DNS Server(s)	`dns-server <ip-1> [ip-2 ... ip-8]`
Set Domain Name	`domain-name <domain-string>`
Set Lease Duration	`lease {<days> [<hours> [<minutes>]] | infinite}`
Set NetBIOS / WINS Server(s)	`netbios-name-server <ip-1> [ip-2]`
Disable DHCP Service	`no service dhcp`
Enable DHCP Service	`service dhcp`
Configure DHCP Relay	`ip helper-address <server-ip>`
View Active Bindings	`show ip dhcp binding [ip-address]`
View Pool Statistics	`show ip dhcp pool [<pool-name>]`
View Server Statistics	`show ip dhcp server statistics`
View Address Conflicts	`show ip dhcp conflict`
Clear a Lease / All Leases	`clear ip dhcp binding {* | <ip-address>}`
Clear Conflict(s)	`clear ip dhcp conflict {* | <ip-address>}`

NOTE — Modes

`ip dhcp excluded-address`, pool creation, and `service dhcp` / `no service dhcp` run in global configuration mode. `network`, `default-router`, `dns-server`, `domain-name`, `lease`, and `netbios-name-server` run inside DHCP pool configuration mode (dhcp-config). `ip helper-address` is applied on the client-facing interface, not globally.

Example — Basic DHCP Pool:

```
R1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.10
R1(config)# ip dhcp pool LAN-POOL
R1(dhcp-config)# network 192.168.10.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.10.1
R1(dhcp-config)# dns-server 8.8.8.8 8.8.4.4
R1(dhcp-config)# lease 7
```

IPV6 DHCP (SERVER, POOL & RELAY)

Create DHCPv6 Pool	`ipv6 dhcp pool <pool-name>`
Set Address Prefix (Stateful)	`address prefix <ipv6-prefix/length> [lifetime ...]`
Set DNS Server(s)	`dns-server <ipv6-address>`
Set Domain Name	`domain-name <domain-string>`
Link Pool to Interface	`ipv6 dhcp server <pool-name>`
Stateless Flag (DNS/Domain Only)	`ipv6 nd other-config-flag`
Stateful Flag (Address + Info)	`ipv6 nd managed-config-flag`

Configure DHCPv6 Relay	`ipv6 dhcp relay destination <ipv6-address>`
View Active Bindings	`show ipv6 dhcp binding`
View Pool Statistics	`show ipv6 dhcp pool [<pool-name>]`
View Interface Setup	`show ipv6 dhcp interface [<interface>]`
Clear Bindings	`clear ipv6 dhcp binding {* | <ipv6-address>}`

NOTE — Modes

`ipv6 dhcp pool` runs in global configuration mode. `address prefix`, `dns-server`, and `domain-name` run inside DHCPv6 pool configuration mode (dhcp-config). `ipv6 dhcp server`, the `ipv6 nd` RA flags, and `ipv6 dhcp relay destination` are all applied on the client-facing interface, not globally.

NOTE — Stateless vs. Stateful

`ipv6 nd other-config-flag` tells clients to use SLAAC for their address and DHCPv6 only for DNS/domain info (stateless DHCPv6). `ipv6 nd managed-config-flag` tells clients to obtain their full address and info from the DHCPv6 server (stateful DHCPv6).

Example — Stateful DHCPv6 Pool:

```
R1(config)# ipv6 dhcp pool LAN-POOL6
R1(config-dhcpv6)# address prefix 2001:DB8:ACAD:10::/64
R1(config-dhcpv6)# dns-server 2001:4860:4860::8888
R1(config-dhcpv6)# domain-name example.com
R1(config)# interface g0/0/1
R1(config-if)# ipv6 dhcp server LAN-POOL6
R1(config-if)# ipv6 nd managed-config-flag
```

CLI SHORTCUTS & NAVIGATION

Tab	`Auto-completes entered CLI command.`
Up / Down Arrows	`Recalls previous command history.`
Ctrl + Shift + 6	`Aborts ping, traceroute, or DNS lookup.`
Ctrl + A / Ctrl + E	`Moves cursor to line start / end.`